

R3 v4 — PRD Achievement Tracker

PRD v3.0 · Build Audit Against Confirmed Monorepo State

audited: 2026-Q1 · build: pnpm monorepo · test coverage: 42 vitest cases

62%

OVERALL PRD COMPLETE

2/5

CORE FEATURES SHIPPED

42

VITEST CASES PASSING

3

CRITICAL GAPS TO MVP

MVP Completion — Investor-Ready Threshold

62%

Current state

▲ 75% = demo-ready

▲ 100% = investor-ready

Shipped & verified

Partially built

Not started

Blocked / missing

SECTION 8 — CORE FEATURES (MVP)

AI Auto-Leveling	11 files · LUFS/RMS analysis · ghost knobs · confidence gating · 20 Vitest cases	✓ 100% Done
Smart Transitions	9 files · 7 types · Camelot wheel harmonic scoring · 22 Vitest cases	✓ 100% Done
Mix Suggestion System	Frequency masking · dynamic imbalance · arrangement conflict triggers	○ Not Started
LLPTE Core Engine	6 packages · named nodes · 10ms inference · node graph UI · WebSocket	~ 90% Done
Time Savings Tracking	Session metrics · AI action log · export PNG/JSON · inline chip	○ Not Started

Evidence: Auto-Leveling in [packages/llpte-ai/](#) (11 files, 20 Vitest). Smart Transitions in [packages/llpte-transition-graph/](#) (9 files, 22 Vitest). LLPTE pipeline confirmed live at [inference 10ms](#) in shipped UI screenshot. Mix Suggestion System + Time Savings: next in build queue per session notes.

SECTION 11 — TECHNICAL INFRASTRUCTURE

pnpm Monorepo	Workspace config · client + server + packages + shared	✓ Done
React + Vite Frontend	client/ · Tailwind · Radix UI · tsconfig confirmed	✓ Done
Express + tRPC + WebSocket	server/ · procedures.ts · ws/ · real-time state sync	✓ Done
Drizzle + PostgreSQL	db/schema/ · drizzle/migrations/ · drizzle.config.ts	✓ Done
WASM + SharedArrayBuffer + AudioWorklet	Real-time audio engine · ≤3ms target per PERF PRD	~ 85% Done
WebGPU Rendering	OffscreenCanvas compute shaders · ≤4ms target · Canvas2D fallback	~ 60% Done
Docker + Railway Deploy	docker-compose.yml · railway.toml · nginx.conf confirmed	✓ Done
MessagePack Binary Protocol	Per PERF PRD · LLPTE signal data transport · ≤50ms round-trip	○ Spec'd Only

SECTION 15 — SECURITY & AUTH

Auth System (3 Bugs)	trpcAuth mount · hydrateFromToken() race · loading state guard — all fixed	✓ Fixed
bcrypt + JWT	bcrypt v6.0.0 lazy-init fixed · JWT jsonwebtoken v9.0.3 · httpOnly cookie	✓ Done
Path Traversal Protection	Upload route — allowlist-validated · audit-confirmed	✓ Fixed
Admin Seeder	Idempotent · server/scripts/ confirmed	✓ Done
Rate Limiting	express-rate-limit v7.5.1 · confirmed in node_modules · rules TBD	~ Partial
WCAG Accessibility	aria-live for ghost knobs · focus rings · reduced-motion guards	○ Not Started

SECTION 7 — UI ARCHITECTURE (FIVE ZONES)

Zone 1 — Top Nav Bar	Transport · BPM · Pro badge · CPU meter · user avatar — all visible in screenshot	✓ Shipped
Zone 2 — Sample Browser	Search · tree · waveform preview · file metadata — confirmed in screenshot	✓ Shipped
Zone 3 — Arrangement / Timeline	8 tracks · playhead · minimap · context menu · AI MIX track · vectorscope	✓ Shipped
Zone 4A — DJ Mixer	6 channel strips · 3-band EQ · VU meters · crossfader · performance pads	✓ Shipped
Zone 4B — LLPTE Node Graph	5 nodes · animated connectors · inference badge · oscilloscope · sparklines	~ 90% Shipped
Zone 5 — Effects Rack	6 modules visible · Reverb, Delay, Compressor, EQ confirmed · full param spec TBD	~ 70% Done

SECTIONS 4, 14, 18, 24–26 — PRODUCT, ANALYTICS, GTM, DEMO

Subscription Tiers (Stripe)	Stripe v20 · LemonSqueezy · tier migration audit-fixed · Pro badge live	~ 65% Done
Analytics & Telemetry Schema	Event schema designed · aiDecisionLog table defined · ingestion pipeline TBD	~ 30% Done
Onboarding Flow	Spec'd in PRD · tooltip sequence · empty state · activation criteria	○ Not Started
PERF PRD (6-Layer Optimization)	R3v4_PERF_PRD.html produced · WASM/SIMD/WebGPU/pool/msgpack/matviews	~ Spec Done, Impl Par
Investor Demo Script	12-minute script · 7 beats · full QA checklist · scripted in PRD §26	Script Ready, Demo N
Go-to-Market & Content Strategy	4 channels defined · hardware partnership math · KPIs set	○ Strategy Only

CRITICAL GAPS — BLOCKING MVP COMPLETION

Gap 1 — Mix Suggestion System

Frequency masking detection, dynamic imbalance alerts, arrangement conflict flagging, and the inline suggestion panel with Accept / Reject / Preview are completely unbuilt. This is the second-most visible AI feature in a demo after Auto-Leveling — its absence means only one AI feature fires during playback.

Estimate: 2–3 weeks · Priority: CRITICAL

Gap 2 — Time Savings Tracking

The session metrics schema is designed but not wired. No session summary panel, no export PNG/JSON, no inline action counter in the top nav. This is the single most important investor demo artifact — the "42% faster" moment. Without it, the demo lacks a quantifiable proof point.

Estimate: 1–2 weeks · Priority: CRITICAL

Gap 3 — Onboarding Flow

No empty state, no tooltip sequence, no activation funnel. New users land in the arrangement view with no guidance. Target 70% activation in first session is unachievable without this. Affects both alpha user conversion and investor perception of product maturity.

Estimate: 1 week · Priority: HIGH

Gap 4 — Analytics Ingestion Pipeline

The event schema and aiDecisionLog table are defined. But without the ingestion layer wired to every AI action, the acceptance rate metric (target $\geq 65\%$) cannot be measured — making it impossible to prove the PRD's core success criterion during due diligence.

Estimate: 1 week · Priority: HIGH

Gap 5 — PERF PRD Implementation

The 6-layer performance contract (WASM $\leq 3\text{ms}$, quantized inference $\leq 2\text{ms}$, WebGPU $\leq 4\text{ms}$, pool allocators, MessagePack, materialized views) is fully specified but partially implemented. The demo requires p50 $\leq 15\text{ms}$ sustained across 8 tracks — this needs a full benchmark run before any investor meeting.

Estimate: 3–4 weeks · Priority: MEDIUM (pre-demo)

Gap 6 — WCAG Accessibility

aria-live for ghost knobs, focus rings, reduced-motion guards, and screen reader labels are unimplemented. Not blocking for demo but required before public launch and any enterprise/label conversation.

Estimate: 1 week · Priority: LOW (post-MVP)

RECOMMENDED BUILD ORDER TO DEMO-READY (75%)

1

Time Savings Tracking

Wire sessionMetrics schema → session summary panel → Export PNG. The "42% faster" moment is non-negotiable for any investor conversation.

1–2 weeks

2

Mix Suggestion System

Frequency masking detection → suggestion panel → inline timeline markers. Makes the AI feel like it's always watching during the demo.

2–3 weeks

3

Analytics Ingestion

Wire every AI action to aiDecisionLog. Enables the acceptance rate metric — the core PRD success criterion.

1 week

4

Onboarding Flow

Empty state → tooltip sequence → first-session activation. Required to convert alpha users without white-glove onboarding calls.

1 week

5

PERF PRD Benchmark Run

Run the 17-item QA checklist. Confirm p50 $\leq 15\text{ms}$ at 8 tracks. Fix any dropout issues. No investor meeting before this passes.

1 week

Bottom Line

The **technical foundation is strong** — the hardest engineering (LLPTE pipeline, audio engine, monorepo architecture, auth, real-time WebSocket, database) is done. The UI is shipped and visually impressive. Two of five

core AI features are fully implemented with 42 Vitest cases passing.

The **missing 38% is primarily product-layer work**, not architecture: Time Savings Tracking, Mix Suggestion System, analytics wiring, and onboarding. These are buildable in **5–7 focused weeks** and would take R3 from 62% to 100% investor-ready.

R3 v4 PRD Achievement Tracker · v3.0 · Confidential

Built: pnpm monorepo · 42 Vitest cases · Confirmed: LLPTE 10ms inference · 847 active edges · Shipped UI verified